STATISTICS & MACHINE LEARNING · LESSON 9 OF 28

Data Preprocessing and Feature Engineering for ML

Prepare features that models can learn from. Imputation strategies, outlier handling, encoding categorical variables (one-hot, ordinal, target encoding), scaling (standard, min-max, robust), transformations (log, Box-Cox/Yeo-Johnson), binning, interactions and polynomial features, date and text features, and feature selection, all inside leak-free pipelines.

ML class 6 · ✦ scikit-learn 1.7+ · Intermediate · ⏱ 120 min

By the end of this lesson you can

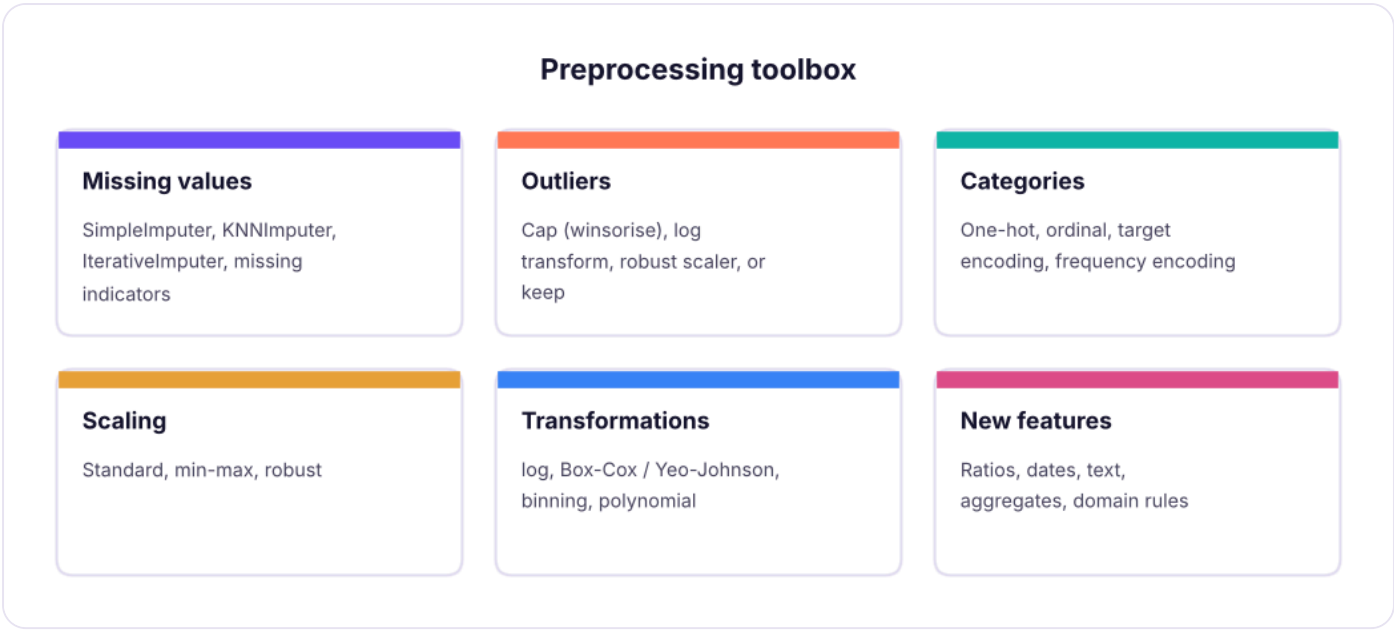
- Impute missing values with simple, group-aware and model-based strategies and add missing indicators
- Encode nominal and ordinal categories correctly, including high-cardinality columns
- Choose a scaler and know which algorithms need scaling
- Engineer features from numbers, dates, text and domain knowledge
- Select features with filter, wrapper and embedded methods without leaking test data

Datasets used

`hr_employees.csv` `house_prices.csv`

Why preprocessing matters

Algorithms only see numbers. How you represent the data often matters more than which algorithm you choose ("better data beats fancier models").



The preprocessing toolbox, applied inside a pipeline.

Missing values

Python

```
import pandas as pd, numpy as np
from sklearn.impute import SimpleImputer, KNNImputer
from sklearn.experimental import enable_iterative_imputer  # noqa: F401
from sklearn.impute import IterativeImputer
h = pd.read_csv("house_prices.csv")
num = h[["area_sqft", "bedrooms", "bathrooms", "age_years", "price_lakh"]]
mask = num.area_sqft.isna(); print("missing area:", mask.sum())
for name, imp in [("median", SimpleImputer(strategy="median")), ("KNN (5 neighbours)",
KNNImputer(n_neighbors=5)), ("iterative (regression)", IterativeImputer(random_state=0))]:
    filled = pd.DataFrame(imp.fit_transform(num), columns=num.columns)
    print(f"{name:24s} imputed areas: mean {filled.loc[mask, 'area_sqft'].mean():7.0f}, sd
{filled.loc[mask, 'area_sqft'].std():6.0f}")
print("SimpleImputer(add_indicator=True) adds a column:", SimpleImputer(strategy="median",
add_indicator=True).fit(num[["area_sqft"]]).get_feature_names_out().tolist())
```

OUTPUT

```
missing area: 25
median                   imputed areas: mean    1240, sd      0
KNN (5 neighbours)       imputed areas: mean    1396, sd    294
iterative (regression)   imputed areas: mean    1384, sd    310
SimpleImputer(add_indicator=True) adds a column: ['area_sqft',
'missingindicator_area_sqft']
```

The median fills every gap with one value (sd 0); KNN and iterative imputers use the other columns (bedrooms, price), so they produce realistic, varied values. In real projects compare strategies with cross-validation, and never use the target to impute features at prediction time.

Encoding categorical variables

Python

```
from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder, TargetEncoder
hr = pd.read_csv("hr_employees.csv")
print(hr.education.unique().tolist(), hr.department.nunique(), "departments |",
hr.job_title.nunique(), "job titles")
oh = OneHotEncoder(handle_unknown="ignore", sparse_output=False).fit(hr[["department"]])
print("one-hot:", oh.get_feature_names_out().tolist())
order = [["Diploma", "Bachelor's", "Master's", "PhD"]]
present = [e for e in order[0] if e in set(hr.education)]
oe = OrdinalEncoder(categories=[present]).fit(hr[["education"]])
print("ordinal:", dict(zip(present, oe.transform(pd.DataFrame({"education":
present})).ravel())))
y = (hr.attrition == "Yes").astype(int)
te = TargetEncoder(target_type="binary", random_state=0).fit(hr[["job_title"]], y)
print("target encoding (smoothed attrition rate per job title, first 4):",
dict(zip(te.categories_[0][:4], te.encodings_[0][:4].round(3))))
print("frequency encoding:",
hr.job_title.map(hr.job_title.value_counts(normalize=True)).round(3).head(3).tolist())
```

OUTPUT

```
["Bachelor's", "Master's", 'Diploma', 'PhD'] 6 departments | 18 job titles
one-hot: ['department_Finance', 'department_HR', 'department_IT',
'department_Operations', 'department_R&D', 'department_Sales']
ordinal: {'Diploma': np.float64(0.0), "Bachelor's": np.float64(1.0), "Master's":
np.float64(2.0), 'PhD': np.float64(3.0)}
target encoding (smoothed attrition rate per job title, first 4): {'Accountant':
np.float64(0.256), 'Analyst': np.float64(0.28), 'Data Scientist': np.float64(0.247),
'Developer': np.float64(0.295)}
frequency encoding: [0.049, 0.063, 0.063]
```

Encoding	For	Notes
One-hot	Nominal, few categories	One column per category; `handle_unknown="ignore"`; `drop="first"` for linear models if needed
Ordinal	Ordered categories (education, size S/M/L)	Give the order explicitly
Target encoding	High-cardinality nominal (job title, city, merchant)	Uses the target: must be cross-fitted (sklearn's `TargetEncoder` does this)
Frequency / count	High cardinality	Simple, leak-free
Hashing, embeddings	Very high cardinality (IDs, text)	Fixed size; embeddings in deep learning

WATCH OUT **Label encoding nominal data**

Mapping cities to 0, 1, 2... tells linear models and KNN that "Pune (2) > Delhi (1)", a false order. Use one-hot encoding for nominal variables (tree models tolerate ordinal codes better).

Scaling

Python

```
from sklearn.preprocessing import StandardScaler, MinMaxScaler, RobustScaler
inc = hr[["monthly_income"]]
out = pd.DataFrame({"raw": inc.monthly_income})
for name, sc in [("standard", StandardScaler()), ("minmax", MinMaxScaler()), ("robust",
RobustScaler())]: out[name] = sc.fit_transform(inc).ravel()
print(out.describe().loc[["mean", "std", "min", "50%", "max"]].round(2))
```

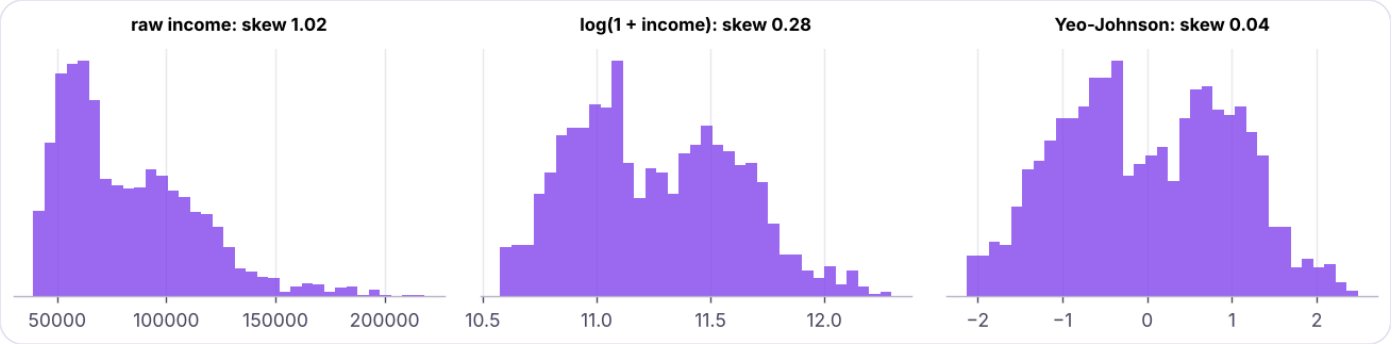
OUTPUT

```
             raw  standard  minmax  robust
mean    82610.67     -0.00    0.24    0.18
std     31145.80      1.00    0.17    0.71
min     39167.00     -1.40    0.00   -0.81
50%     74708.50     -0.25    0.20    0.00
max    218417.00      4.36    1.00    3.28
```

Scaler	Result	Use
StandardScaler	mean 0, sd 1	Default for linear models, SVM, PCA, neural networks
MinMaxScaler	range 0–1	Bounded inputs (neural networks, images)
RobustScaler	median 0, IQR 1	Data with outliers

Needs scaling: KNN, K-means, SVM, PCA, regularised linear/logistic regression, neural networks, gradient descent. **Does not:** decision trees, random forests, gradient boosting (they split on thresholds).

Transformations



Log and Yeo-Johnson transforms make right-skewed income closer to symmetric.

Log transforms help linear models with multiplicative effects (prices, incomes) and can be applied to the **target** too (`TransformedTargetRegressor`). Binning (`KBinsDiscretizer` , `pd.cut`) trades detail for robustness and interpretability.

Feature engineering

Good features encode domain knowledge the algorithm cannot easily find.

Python

```python
hr["hire_date"] = pd.to_datetime(hr.hire_date)
fe = pd.DataFrame({
    "income_per_year_tenure": hr.monthly_income / (hr.years_at_company + 1),       # ratio
    "is_long_commute": (hr.distance_from_home_km > 20).astype(int),                #
threshold from domain knowledge
    "overtime_x_low_sat": ((hr.overtime == "Yes") & (hr.job_satisfaction ⩽
2)).astype(int),   # interaction
    "hire_month": hr.hire_date.dt.month, "hired_in_q4":
hr.hire_date.dt.quarter.eq(4).astype(int),   # date parts
    "income_vs_dept_median": hr.monthly_income /
hr.groupby("department").monthly_income.transform("median"),   # group-relative
    "title_is_manager": hr.job_title.str.contains("Manager").astype(int),          # text
pattern
})
fe["left"] = (hr.attrition == "Yes").astype(int)
print(fe.corr()["left"].drop("left").round(3).sort_values())
print(fe.groupby("overtime_x_low_sat").left.mean().round(3).to_dict())
```

OUTPUT

```
income_vs_dept_median    -0.109
title_is_manager         -0.048
hire_month                0.038
hired_in_q4               0.043
income_per_year_tenure    0.057
is_long_commute           0.094
overtime_x_low_sat        0.165
Name: left, dtype: float64
{0: 0.229, 1: 0.43}
```

Source	Feature ideas
Numbers	Ratios, differences, logs, bins, polynomial and interaction terms
Dates	Day of week, month, holiday flags, days since last event, tenure, seasonality (sin/cos)
Transactions	Recency, frequency, monetary (RFM), averages over 7/30/90 days, trends
Text	Length, keyword flags, TF-IDF, embeddings (lesson 25 and GenAI track)
Geography	Distance to centre/store, region aggregates
Groups	Value relative to group mean/median (income vs department)

Python

```
from sklearn.preprocessing import PolynomialFeatures
pf = PolynomialFeatures(degree=2, include_bias=False).fit(pd.DataFrame({"area": [1], "age": [1]}))
print(pf.get_feature_names_out().tolist())
```

OUTPUT

```
['area', 'age', 'area^2', 'area age', 'age^2']
```

Feature selection

More features are not always better: irrelevant ones add noise, cost and overfitting.

Python

```
from sklearn.feature_selection import mutual_info_classif, SelectKBest, RFE
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
Xn = pd.get_dummies(hr[["age", "monthly_income", "years_at_company", "job_satisfaction", "distance_from_home_km", "performance_rating", "overtime", "department"]], drop_first=True).astype(float)
mi = pd.Series(mutual_info_classif(Xn, y, random_state=0), index=Xn.columns).sort_values(ascending=False)
print("filter (mutual information):", mi.round(3).head(5).to_dict())
rfe = RFE(LogisticRegression(max_iter=2000), n_features_to_select=4).fit(StandardScaler().fit_transform(Xn), y)
print("wrapper (RFE, logistic):", Xn.columns[rfe.support_].tolist())
rf = RandomForestClassifier(300, random_state=0, min_samples_leaf=5).fit(Xn, y)
print("embedded (random forest importance):", pd.Series(rf.feature_importances_, Xn.columns).sort_values(ascending=False).round(3).head(5).to_dict())
```

OUTPUT

```
filter (mutual information): {'years_at_company': 0.025, 'job_satisfaction': 0.019, 'department_HR': 0.017, 'monthly_income': 0.009, 'overtime_Yes': 0.007}
wrapper (RFE, logistic): ['years_at_company', 'job_satisfaction', 'distance_from_home_km', 'overtime_Yes']
embedded (random forest importance): {'monthly_income': 0.242, 'distance_from_home_km': 0.18, 'age': 0.155, 'years_at_company': 0.117, 'job_satisfaction': 0.108}
```

The three methods disagree. That is normal: they measure different things. Random-forest (impurity) importances also favour continuous features with many distinct values, so check them with permutation importance (lesson 27).

Method	How	Pros / cons
Filter	Score each feature (correlation, chi-square, mutual information)	Fast; ignores interactions
Wrapper	Search subsets with a model (RFE, forward selection)	Strong; slow, risk of overfitting
Embedded	Selection during training (Lasso L1, tree importances)	Efficient; model-specific

Do feature selection **inside** cross-validation (in the pipeline) so the test data never influences which features are chosen.

Putting it together: a leak-free pipeline

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline, make_pipeline
from sklearn.model_selection import cross_val_score
X = hr.drop(columns=["emp_id", "attrition", "hire_date"])
pre = ColumnTransformer([
    ("num", make_pipeline(SimpleImputer(strategy="median"), StandardScaler()), ["age",
"years_at_company", "monthly_income", "job_satisfaction", "distance_from_home_km",
"performance_rating"]),
    ("ord", OrdinalEncoder(categories=[present]), ["education"]),
    ("ohe", OneHotEncoder(handle_unknown="ignore"), ["gender", "department", "overtime",
"city"]),
    ("te", TargetEncoder(target_type="binary", random_state=0), ["job_title"]),
])
pipe = Pipeline([("pre", pre), ("sel", SelectKBest(mutual_info_classif, k=12)), ("model",
LogisticRegression(max_iter=2000))])
scores = cross_val_score(pipe, X, y, cv=5, scoring="roc_auc")
print(f"5-fold ROC-AUC {scores.mean():.3f} ± {scores.std():.3f}")
```

OUTPUT

```
5-fold ROC-AUC 0.661 ± 0.013
```

Summary

KEY TAKEAWAYS

- Impute with medians/modes, KNN or iterative imputers, and add missing-value indicators when missingness carries information.
- One-hot for nominal, ordinal encoding with an explicit order, target or frequency encoding for high-cardinality columns.
- Scale for distance- and gradient-based models (standard, min-max, robust); trees do not need scaling.
- Transform skewed variables (log, Yeo-Johnson); engineer ratios, interactions, date and group-relative features from domain knowledge.
- Select features with filter, wrapper or embedded methods, always inside the cross-validated pipeline.

Quick quiz

6 questions · tap an answer to check it

1. Which encoding suits "education level" (Diploma < Bachelor's < Master's < PhD)?

A Ordinal encoding with the order given explicitly

B One-hot only

C Target encoding only

D Leave as text

Answer: A. The categories have a natural order.

2. Which algorithm does NOT need feature scaling?

A KNN

B SVM

C Random forest

D K-means

Answer: C. Trees split on thresholds, unaffected by scale.

3. Why must target encoding be cross-fitted?

A Using a row's own target to encode it leaks the answer and overfits

B It is slow otherwise

C It needs scaling

D It only works on numbers

Answer: A. sklearn's TargetEncoder handles this internally.

4. What does RobustScaler use?

A Mean and standard deviation

B Median and interquartile range

C Min and max

D Logarithms

Answer: B. Robust to outliers.

5. Where should feature selection happen?

A On the full dataset before splitting

B Inside the pipeline, within cross-validation on training data

C On the test set

D After deployment

Answer: B. Otherwise test information leaks into the choice.

6. Which is an interaction feature?

A monthly_income

B overtime AND low job satisfaction

C log(income)

D hire_month

Answer: B. It combines two features.

EXERCISE

Practice (house_prices.csv)

(1) Engineer `price_per_sqft`, `rooms = bedrooms + bathrooms`, `is_new = age_years ⩽ 5`. Which correlate most with price? (Careful: price_per_sqft uses the target, so it cannot be a feature!) (2) Build a pipeline with KNN imputation, one-hot city, standard scaling and Ridge regression; report 5-fold R^2. (3) Try a log-transformed target with `TransformedTargetRegressor`.

Show solution (worked answer)

Python

```
from sklearn.linear_model import Ridge
from sklearn.compose import TransformedTargetRegressor
h["rooms"] = h.bedrooms + h.bathrooms; h["is_new"] = (h.age_years ⩽ 5).astype(int)
Xh = h.drop(columns=["house_id", "price_lakh"]); yh = h.price_lakh
pre_h = ColumnTransformer([("num", make_pipeline(KNNImputer(), StandardScaler()),
["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km",
"parking", "rooms", "is_new"]), ("cat", OneHotEncoder(handle_unknown="ignore"),
["city"])])
print(cross_val_score(Pipeline([("p", pre_h), ("m", Ridge())]), Xh, yh, cv=5,
scoring="r2").mean())
print(cross_val_score(Pipeline([("p", pre_h), ("m",
TransformedTargetRegressor(Ridge(), func=np.log1p, inverse_func=np.expm1))]), Xh, yh,
cv=5, scoring="r2").mean())
```

INTERVIEW CORNER

Questions you may be asked

Q: How do you handle categorical variables with 1,000 levels? Target encoding (cross-fitted), frequency encoding, grouping rare levels into "Other", hashing, or learned embeddings; tree-based libraries like CatBoost handle them natively.

Q: Normalisation vs standardisation? Normalisation (min-max) rescales to 0–1; standardisation centres to mean 0 and sd 1. Use standardisation by default; min-max for bounded inputs.

Q: Give an example of a feature you engineered that helped. Use STAR: e.g. "days since last purchase" and "spend in the last 30 days vs previous 30" for churn, which raised recall in the top decile from X to Y.

Practise more online

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